Knowledge and Language 2025/2026

Masters in Informatics Engineering
Natural Language Processing and Knowledge Representation

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1 Problem Statement

General LLMs can answer many questions but they can lack niche or culturally specific Knowledge. The challenge we are trying to tackle is building a system that can reliably and accurately answer complex, domain-specific questions about Portuguese cuisine and other recipes using a variety of datasets.

2 Project Goal

The goal of the project is to develop a ChatBot that leverages a built Structured knowledge Graph of food recipes from the datasets. The ChatBot is then able to query the knowledge graph from a Natural Language prompt and provide accurate answers to the user.

3 Data and Tools and Proposed Approach

The datasets are a Dataset of Portuguese recipes taken from cooking books that has the ingredients and region of origin of each recipe and a more complete dataset of recipes from Kaggle that has ingredients, `prep_time`, `cook_time`, directions etc. The system relies on a curated dataset of recipes that has been transformed into a structured knowledge graph using the RDF (Resource Description Framework) model. Each RDF triplet has the form:

- subject (it can be a specific recipe or ingredient, e.g., "Bacalhau à Brás"),
- predicate (it can be a variety of attributes like hasIngredient, hasPrepTime)
- object related (ex tomato, 15 minutes)

this allows for querying relationships between recipes, ingredients, and other attributes. The knowledge base is accessed through SPARQL, which allows precise retrieval of information based on user intents and entities extracted using NLP tools like spaCy and BERT. A Retrieval-Augmented Generation (RAG) setup interprets the SPARQL templates and generates context-aware responses. The final answer is then returned to the user via a React front-end, which provides an interface for interacting with the chatbot.

4 Conection with the reviewed literature (onde encontramos artigos q dão ideias boas nestes topicos)

Knowledge Graph Construction/Enrichment:
Knowledge Retrieval via NLP:
SPARQL Query Generation:

Retrieval-Augmented Generation (RAG):

5 checkpoint dates

Checkpoint 1 -¿ 18 November 2025 Checkpoint 2 -¿ 2 December 2025